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Decentralization, Service Delivery, and Legitimacy

The Case of Mali

Gaetan Mertens
Nouhoum Touré

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Abstract

In fragile, conflict-affected, and violent contexts, decentralization is widely promoted to improve service delivery and strengthen the social contract by shifting power closer to citizens. However, empirical evidence on whether these reforms deliver in environments marked by weak state capacity and low institutional trust remains limited. This paper evaluates Mali's Project for the Deployment of State Resources for the Improvement of Services program, which implemented performance-based conditional grants across 102 communes between 2021 and 2024. Drawing on administrative budget data and georeferenced Afrobarometer survey data, the study assesses the impacts of performance-based conditional grants on local fiscal performance, service delivery, and institutional legitimacy. The results show that a 1 percent increase in performance-based conditional grant allocations is associated with a 1.2 percent increase in investment expenditures and measurable improvements in budget balances. Expenditures were reallocated toward social sectors (education, health, water, and agriculture), consistent with program objectives. A difference-in-differences analysis points to statistically significant increases in citizen satisfaction with public services, particularly health (+2.6 percentage points), water and sanitation (+1.4), and education (+1.0), as well as an 18.4-point rise in trust in municipal councils. Institutional "spillover" effects extended to the national level, with trust in the presidency and parliament increasing by 9.1 and 7.3 points, respectively. A scenario-based cost-effectiveness analysis points to potentially substantial returns under the stated assumptions. Illustrative estimates suggest that observed trust gains could be associated with USD 57 million to USD 141 million in additional annual tax revenues, equivalent to 1.1 to 2.8 times the program's total cost, through enhanced voluntary compliance—assuming a transmission of trust into compliance broadly consistent with parameters in the literature. These findings are consistent with the view that well-designed performance-based transfers may improve service outcomes and contribute to rebuilding public trust in fragile, conflict-affected, and violent contexts, with effects that appear to extend beyond the local level. Policy implications include considering institutionalization of the performance-based conditional grants within Mali's fiscal architecture and the design of a medium-term roadmap for sustainability and possible scale-up.

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Decentralization, Service Delivery, and Legitimacy: The Case of Mali*

Gaetan Mertens† and Nouhoum Touré‡

Authorized for distribution by Gael Raballand, Country Manager for Tajikistan and
Jens Kromann Kristensen, Practice Manager, Prosperity Vertical, World Bank Group

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†World Bank. email: gmertens@worldbank.org.

‡CES, Université Paris 1 Panthéon Sorbonne. email: nouhoum.toure@univ-paris1.fr.

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1 Introduction

In fragility, conflict-affected and violent (FCV) contexts, decentralization is often promoted as a strategic approach to improve the delivery of public services and to strengthen the social contract by bringing power and resources closer to citizens. However, empirical evidence on the actual effectiveness of such reforms in environments characterized by weak state capacity and a structural deficit of institutional trust remains limited. This tension between normative ambition and institutional reality raises a fundamental question: do decentralization reforms truly deliver on their promises in the most fragile settings?

Decentralization has, for several decades, been considered a powerful institutional response to the challenges of governance, public resource allocation, and local development. In developing countries, where central governments are often distant from territorial realities and where public services are underfunded or poorly adapted, decentralization offers the promise of a reconfiguration of roles, responsibilities, and resources among different levels of government. The classic theoretical argument, introduced by Oates (1972), posits that local governments are better positioned to understand local preferences and adjust the provision of public goods accordingly. This "better fit" hypothesis is grounded in the principle of subsidiarity, which holds that any public function should be carried out by the level of government closest to the citizen that is capable of performing it effectively.

However, empirical evidence shows that the effects of decentralization on public service delivery are highly variable and heavily dependent on institutional context. Numerous studies have emphasized that the quality of local governance, the technical capacity of territorial administrations, and the presence of accountability mechanisms are decisive factors. For instance, Bardhan and Mookherjee (2006) demonstrate that in countries with weak local institutions, decentralization can lead to resource capture by local elites, increased interregional inequalities, and reduced efficiency. This concern is further supported by the work of Prud'Homme (1995), who warns of the risks of macroeconomic incoherence, fragmentation of public policies, and poor resource allocation when responsibilities are transferred without adequate calibration. In sub-Saharan Africa, several documented experiences have tested these theories. In Ghana, for example, the allocation of conditional resources to districts led to measurable improvements in access to healthcare and education, but only in areas where administrative capacity was sufficient to plan and implement projects (Oppong, 2020, Mogues and Benin, 2012). In Bolivia, Faguet (2004) observed that massive resource transfers to municipalities produced significant social investment outcomes, although the uneven distribution of local capacity also widened performance gaps between municipalities. These experiences show that the beneficial effects of decentralization are not automatic; they depend on a coherent alignment between local autonomy and mechanisms for control, incentives, and support. Ultimately, decentralization remains a conditional promise: it can contribute to

local development and improve public service delivery only when it is anchored in strong local institutions, appropriate incentives, and effective implementation capacity.

In the specific context of Mali, marked by a protracted security crisis spanning several decades, rapid population growth, and significant administrative fragmentation, the effective implementation of decentralized policies requires more than just the transfer of resources. It calls for a thorough overhaul of local governance arrangements. The Project for the Deployment of State Resources for the Improvement of Services (PDREAS) thus represents a structured attempt to reconcile national political ambitions with complex operational realities on the ground, by combining performance incentives, institutional support, and territorial coordination.

This study aims to assess the extent to which the PDREAS program has contributed to strengthening local governance, improving budget management in communes, and expanding access to essential services. More specifically, it examines the impact of its core mechanism, the Performance-Based Conditional Grant (Dotation Conditionnelle de Performance, DCP), along three key dimensions: local budgetary performance; the sectoral allocation of public expenditures, particularly in education, health, and economic resilience; and citizens' perceptions of institutional legitimacy at both the local and national levels. Additionally, the study conducts a comprehensive cost-effectiveness analysis to assess the fiscal sustainability and potential returns of the program, examining both direct budgetary impacts and indirect effects through enhanced institutional trust and tax compliance.

To achieve these objectives, the study draws on two distinct datasets. First, it uses an original set of administrative budget data covering the 102 communes that benefited from the PDREAS program between 2021 and 2024. These data, extracted from management accounts compiled by the Treasury, allow for a detailed reconstruction of local financial flows, clearly distinguishing between tax revenues, transfers (including the DCP), and executed expenditures disaggregated by type (operating and investment) and by functional sector (education, health, agriculture, water, and infrastructure). The temporal, sectoral, and spatial granularity of this dataset makes it a rare and valuable resource in the African context, enabling in-depth analysis of budgetary priorities and local absorption capacity. Second, the study relies on georeferenced survey data from Afrobarometer, covering the period from 2008 to 2025. These data offer detailed insights into citizens' perceptions regarding the quality of public services, the performance of local authorities, and trust in national institutions. Using GPS coordinates for enumeration areas (EAs), each respondent is matched to a specific commune and assigned an exposure status to the PDREAS program. This matching strategy enables a comparative analysis of changes in citizen perceptions in beneficiary (treated) versus non-beneficiary (non-treated) communes, before and after program implementation.

On the empirical side, two complementary strategies are employed. For the administrative data, a fixed-effects panel regression is used to model the relationship between the amount of DCP

received and several indicators of budgetary performance, including investment levels, budget balance, execution rates, and the sectoral orientation of expenditures. For the Afrobarometer data, a difference-in-differences (DiD) approach is applied. By linking the PDREAS implementation timeline to each commune, this strategy allows for the estimation of the program's differential impact on citizen satisfaction with public services and institutional trust. Individual-level control variables (age, gender, education, employment status) as well as contextual controls (levels of violence based on ACLED data) are included, and standard errors are clustered at the commune level.

The results point to several encouraging patterns associated with the DCP in Mali. On the budgetary side, each marginal increase in DCP allocations is associated with approximately a 1.2% rise in local investment expenditures (point estimate of 1.24), a notable improvement in the budget balance (elasticity of 2.26), a modest increase in own-source tax revenues (elasticity of 0.20), and a higher overall budget execution rate (coefficient of 0.8). Furthermore, local budget allocations are substantially reoriented to priority sectors such as education, health, and food security. With regard to political and institutional legitimacy, Afrobarometer data show a significant increase in citizens' satisfaction with key public services following the program: satisfaction with health services rose by 2.6 percentage points, water and sanitation by 1.4 points, and education by 1 point. In parallel, trust in institutions strengthened considerably, both locally (municipal councils: +18.4 points) and nationally (presidency: +9.1 points; national assembly: +7.3 points), highlighting substantial indirect effects at the national level.

The results regarding trust are particularly important, as trust in institutions is widely recognized as a key enabler of positive civic behaviors, an essential dimension in fragile contexts like Mali. Higher institutional trust has been shown to foster stronger citizen participation in local governance (Putnam et al., 1994), including involvement in planning, budgeting, and oversight processes that enhance accountability and service quality. It also increases openness to contributing to local taxation and service co-financing (Besley, 2020, Besley and Persson, 2014, Sacks, 2012), which is critical for strengthening the fiscal autonomy and sustainability of local governments beyond donor support. Furthermore, trust plays a vital role in conflict mitigation: by encouraging reliance on institutional mechanisms to resolve disputes, higher trust reduces the propensity for informal or violent conflict (Blair, 2019, Lierl, 2021), thereby contributing to social cohesion. The substantial trust gains observed in PDREAS communes (+18.4 points at the local level, with notable spillovers at the national level: +9.1 points for the presidency and +7.3 points for the national assembly) suggest that well-designed performance-based transfers can help activate this virtuous cycle of engagement, cooperation, and stability, reinforcing the social contract and strengthening the foundations for more resilient and inclusive governance. This is particularly relevant in conflict-affected countries like Mali, where chronic institutional fragility and low state legitimacy have historically fueled cycles of violence.

Moreover, a scenario-based cost-effectiveness analysis suggests that these institutional improvements may translate into potentially substantial fiscal returns under stated assumptions. Illustrative estimates suggest that the observed trust gains could be associated with with USD 57 million to USD 141 million annually in additional tax revenues through enhanced voluntary compliance, representing 1.1 to 2.8 times the program's total cost and pointing to the potential for performance-based governance programming to make a meaningful contribution to fiscal sustainability.

These results provide supportive evidence for considering the expansion and institutionalization of the PDREAS model. One of the program's major structural achievements is the adoption of the 2023 Local Government Code (Law No. 2023-004), which establishes a legal and strategic foundation for integrating the DCP mechanism into Mali's intergovernmental fiscal transfer architecture. Anchoring the program within this new framework could play an important role to ensure the predictability, transparency, and long-term sustainability of performance-based transfers. The General Directorate of Local Authorities (DGCT) has expressed the need for technical support in defining appropriate performance criteria.

Nonetheless, several constraints continue to limit the full impact of the program: (i) disbursement delays, particularly in 2021, hindered the timely implementation of communal projects;[1] (ii) information systems do not allow for disaggregated data on the profiles of local elected officials; (iii) inconsistencies between administrative and management accounts undermine the reliability of certain budget analyses; and (iv) the heterogeneity of budget nomenclatures complicates the classification and comparison of expenditures across communes.

A number of options could be considered by authorities and partners to address these limitations and support the program's effectiveness, including: (1) streamlining disbursement procedures to reduce administrative delays; (2) developing a more harmonized budget framework across communes; (3) expanding the systematic collection of data on local elected officials (gender, education level, political affiliation); (4) enriching structural data on communes, including local development plans (PDESC), social infrastructure, and demographic indicators; and (5) institutionalizing citizen perception surveys to monitor outcomes over time. The relative priority and feasibility of these options would depend on the political and administrative context.

In conclusion, PDREAS represents an ambitious and promising experiment in performance-based territorial governance. In this context of multidimensional crisis, the evidence is consistent with the view that an incentive-based fiscal instrument may generate tangible outcomes and contribute to strengthening trust in public action. Considering its institutionalization appears warranted as part of the broader effort to make decentralization an effective lever for development

1. Indeed, the launch of PDREAS coincided with a period of political instability, including two coups d'état in 2020 and 2021. While the program was not formally interrupted, these events contributed to delays in disbursements and affected the timeline of project implementation in several communes.

and stabilization in Mali. The lessons learned from this pilot phase should inform the design of future territorial programs by combining political ambition, accountability requirements, and citizen proximity.

2 Background

Empirical evidence on the effectiveness of decentralization reforms in contexts marked by weak state capacity and low institutional trust remains scarce. Bratton (2012), in a comparative study of twenty African countries including Mali, finds that decentralization, in the absence of robust institutional frameworks, does not necessarily lead to enhanced legitimacy or significant improvements in the quality of local institutions. In other words, the mere devolution of power to local authorities is insufficient to produce visible gains in citizen-perceived governance outcomes. By contrast, our study provides evidence consistent with the view that a structured incentive-based framework, such as the Performance-Based Conditional Grant (DCP), may help activate a virtuous cycle linking improved budgetary performance, targeted social investment, and enhanced institutional legitimacy. Rather than relying solely on participatory mechanisms, this approach engages citizens through the delivery of visible, concrete improvements in their immediate environment, thereby strengthening their trust in both local and national authorities. Moreover, unlike previous studies conducted in more institutionally stable settings, our analysis is situated in the extreme case of a country experiencing chronic fragility for over half a century, compounded by persistent conflict, fragmented governance, and high dependency on external funding. In this respect, our findings complement those of Masaki (2018), who shows that intergovernmental transfers in Tanzania can stimulate local revenue mobilization when paired with adequate capacity building. Our study goes a step further by suggesting that, even in a more degraded institutional context, performance-based conditional grants may not only support improvements in local budget execution but also be associated with potentially substantial fiscal returns and measurable gains in political legitimacy. Specifically, scenario-based estimates suggest that trust gains observed under PDREAS could translate into USD 57 million to USD 141 million in additional tax revenues annually, representing 1.1 to 2.8 times the total program cost. This suggests that trust-building interventions may yield potentially significant state capacity dividends under stated assumptions, even in fragile environments.

It also stands apart from the pioneering analyses of Faguet (2004), who showed that in Bolivia, decentralization led to a significant increase in social investment, while simultaneously exacerbating performance inequalities between municipalities due to uneven local capacities. Moreover, it extends the work of Weingast (2009), who emphasized the importance of institutional incentives and fiscal discipline in making decentralization an effective tool for governance. Our study provides empirical validation of these principles in a low-capacity environment by showing that

communes reallocate their resources toward social sectors while simultaneously improving their fiscal balance.

Furthermore, this study is also distinctive in its methodological approach, combining highly granular administrative data with citizen perception data from Afrobarometer, georeferenced at the commune level. This original combination allows for the documentation of the transmission channels through which conditional grants influence institutional behavior and public opinion. In essence, it provides one of the first evaluations of the impacts of performance-based decentralization in an FCV (fragility, conflict-affected, and violent) context, offering a novel contribution both to theoretical and academic debates and to the formulation of evidence-based public policies. By empirically linking incentive-based financing, improved resource allocation, and the strengthening of institutional legitimacy and trust, the study sheds light on the conditions under which decentralization can serve as a credible lever for state reconstruction and institutional stabilization in the most fragile environments.

The remainder of this study is structured as follows. Section 3 presents the context and implementation modalities of the PDREAS program; Section 4 details the budgetary and perception-based results; Section 5 presents a cost-effectiveness analysis of the program, quantifying fiscal returns from increased institutional trust and assessing the sustainability of the intervention. Section 6 provides some policy recommendations, and concludes this paper. Finally, appendices 7.2 and 7.3 provide additional details on the data sources and the empirical strategy used in this analysis.

3 Context of the PDREAS Program and General Overview

In Mali, decentralization has been a strategic pillar of state reform since the 1990s, supported by an ambitious legal framework, but its implementation remains incomplete. Enshrined in the 1992 Constitution, Malian decentralization is based on the principle of a unitary yet highly deconcentrated state in which territorial collectivities—communes, cercles, and regions—possess legal personality and autonomous management authority. Foundational instruments such as the Decentralization Charter (1993) and the creation of the National Support Fund for Territorial Collectivities (FNACT) laid the groundwork for more autonomous local governance, aimed at bringing administration closer to citizens, strengthening local accountability, and reducing territorial disparities. However, despite this legal architecture, concrete outcomes have been limited. In 2020, actual budget transfers to local governments accounted for just 23% of public expenditures, far below the 30% target set by the 2015 Algiers Peace Agreement. This funding shortfall reflects the persistent gap between normative ambitions and the operational reality of the system.

The implementation of decentralization faces an increasingly complex environment, marked by security instability, institutional fragility, and limited technical capacity among local entities. Since

the political and security crisis of 2012, some regions in the Center and North have partially escaped the control of the central government. In these areas, territorial collectivities often constitute the only visible institutional actors, but their ability to fulfill their functions is severely compromised. Access to state resources is irregular, technical teams are under-resourced, intersectoral coordination mechanisms are underdeveloped, and citizens' trust in local institutions is weakened. This context makes it all the more urgent to deploy a local governance support mechanism that goes beyond mere budget transfers. Such a mechanism should introduce performance-based incentives, strengthen transparency, and foster coordination anchored in local realities.

It is precisely within this perspective that the Project for the Deployment of State Resources for the Improvement of Services (PDREAS) was launched in 2019 with support from the World Bank. Building on lessons learned from the Support Program for Urban Communes in Mali (PACUM), PDREAS represents an effort to scale up results-based financing. Its objective is to strengthen the capacity of local governments to plan, implement, and monitor their investments, while improving the quality of basic public services in key sectors such as education, health, drinking water, and rural infrastructure. The program was initially funded with USD 50 million (approximately 29.2 billion CFA francs), with an additional USD 44.8 million (about 26.18 billion CFA francs) expected from other partners (the Netherlands and the Swiss Cooperation). The launch and early implementation of PDREAS coincided with a period of political instability, including two coups d'état in 2020 and 2021. While the program was not interrupted, these events likely contributed to administrative delays and challenges in fund disbursement. The project is managed by the Ministry of Territorial Administration through a dedicated coordination unit, in partnership with the Regional Directorates of Planning, Statistics, and Information Technology for Land Use Planning and Population (DRPSIAP), the General Directorate of Local Governments (DGCT), and the National Agency for Investment in Territorial Collectivities (ANICT), which also channels other intergovernmental transfers such as FNACT and sectoral funds. In parallel, communes continued to receive other fiscal transfers, notably through ANICT and the FNACT, which primarily financed recurrent expenditures such as salaries and operational costs, while the DCP focused on performance-based capital investments.

The PDREAS program is structured around three complementary components that reflect an integrated approach combining incentives, support, and governance. The first component involves the allocation of a Performance-Based Conditional Grant (DCP) to eligible communes, based on an annual performance assessment. This grant is used to directly finance investment projects. The second component focuses on strengthening local institutions and includes training for communal staff, the deployment of digital tools (ReCoDe[2] for accounting and GEMs[3] for investment geoloca-

2. ReCoDe is a suite of applications and tools for planning, mobilizing, and managing the resources of decentralized local governments. It includes modules for budgetary and inventory accounting, local taxation (TDDRL), and the management of market infrastructure. See www.recode-mali.org.

3. To establish a remote supervision and monitoring evaluation system using KoboToolbox.

tion), support for the implementation of planning frameworks (PDSEC, PTI), and the promotion of citizen participation. Finally, the third component addresses project management, monitoring and evaluation, communication, inter-institutional coordination, and the documentation of lessons learned.

The originality of the PDREAS lies in its Performance-Based Conditional Grant (DCP) mechanism, designed as an incentive tool based on the actual performance of communes. Each year, local governments are assessed against 14 criteria grouped into two categories : five Minimum Mandatory Conditions (CMOs) and nine Performance Criteria (PCs). The CMOs include basic requirements such as the availability of development plans (PDESC), the production of compliant budget documents, and the existence of a participatory framework. The PCs go further, evaluating the quality of budgetary and accounting management, the quality of public service delivery,[4] the use of management tools (BAR, PEFA, ReCoDe), and compliance with principles of transparency and accountability. The evaluation is conducted by the DGCT with support from the DRPSIAP and is validated by the National Committee for Local Finance (CNFL). Only communes that meet all the CMOs are eligible for a DCP.

The allocation of DCP funds follows a structured formula that balances performance-based incentives with territorial equity principles by combining two allocation criteria: a relative performance score and a normalized investment capacity ratio. The DCP allocation for municipality i in year t is given by:

$$DCP_{i,t} = F_t \times \left(\frac{Score_i}{\sum_{j \in \Omega} Score_j} \right) \times \left(\frac{FNACT_Invest_i}{\sum_{j \in \Omega} FNACT_Invest_j} \right)$$

where F_t is the total performance grant envelope for the year, $Score_i$ is the verified performance score of municipality i (ranging from 0 to 100), $FNACT_Invest_i$ is its baseline allocation under the state-financed investment component of the national fund for territorial collectivities, and Ω is the set of municipalities that have fulfilled all five minimum access conditions.

Performance scores are derived from nine indicators grouped under four categories: **Internal Governance** (30 points), including regular council sessions and staff salary payments; **Project Management** (20 points), including existence of a three-year investment plan and investment execution rates; **Transparency and Citizen Participation** (30 points), covering budget consultations, public restitution sessions, and citizen oversight mechanisms; and **Service Delivery Quality** (20 points), including timely school openings and basic health service readiness.

The formula creates a multiplicative incentive system with distinct behavioral implications. The first ratio, $\frac{Score_i}{\sum_{j \in \Omega} Score_j}$, constitutes the performance-based component that rewards better-performing municipalities by assigning them larger shares of the total envelope. This term generates pure

4. For example, in the education sector, a commune is considered to be performing well if, within 30 days of the start of the school year, at least 50% of classrooms are open, one teacher is available for every two classes, and teaching materials are present in schools.

incentive effects, encouraging all eligible communes to improve governance, transparency, and service delivery. The second ratio, $\frac{FNACT_Invest_i}{\sum_{j\in\Omega} FNACT_Invest_j}$, serves as an equity-preserving scaling factor. It ensures that DCP allocations remain proportional to municipalities' baseline fiscal capacity as determined by the national investment formula; which itself incorporates population size, local revenue potential, remoteness, and poverty levels. This prevents large communes from capturing excessive gains while allowing smaller or disadvantaged municipalities to benefit from equitable percentage increases in investment budgets.

The DCP mechanism was applied to a selected group of 102 communes, representing approximately 13% of Mali's 750 communes. As illustrated in Figure 1, a majority of participating communes are in the country's North and Center, areas marked by limited state presence, institutional fragility, and high exposure to conflict and insecurity. In contrast, the district of Bamako, the country's wealthiest and best-equipped urban area, was excluded from the program. Likewise, only a small number of communes from the regions of Kayes and Sikasso were included, despite their demographic weight. These two regions are generally considered among the most economically advantaged in the country, benefiting from stronger financial inflows, especially remittances in Kayes and agricultural surpluses in Sikasso, which partly mitigate local infrastructure and service delivery challenges.[5]

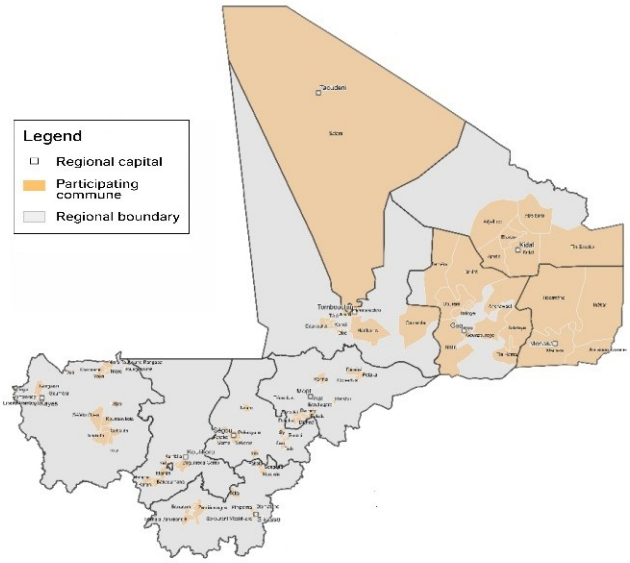


Figure 1: Geographic Distribution of the 102 Communes Covered by the PDREAS Program

5. Programs based on performance-based grants can sometimes risk increasing regional inequalities, as wealthier or better-equipped localities are often more likely to meet performance criteria. However, given the geographic distribution of participating communes, we believe the program's design contributed more to reducing geographic inequalities than reinforcing them. The participating communes were, for the most part, fragile territories in conflict-affected regions, where institutional capacity and access to basic services remain particularly weak.

Today, PDREAS stands not only as an operational investment program but also as a laboratory for territorial reform in Mali. Between 2021 and 2024, nearly 18 billion CFA francs (≈ 32 million USD) were transferred to the 102 beneficiary communes at the time of the study.[6] Table presents descriptive statistics on the DCP amounts allocated to communes, by year and in total. These figures confirm the program's steady scale-up. After a modest start in 2021, with 2.73 billion CFA (≈ 4.8 million USD) disbursed, and a gap year in 2022 due to administrative delays, disbursements accelerated significantly in 2023 and 2024, reaching 7.88 (≈ 14 million USD) and 7.4 billion CFA (≈ 13 million USD) respectively. Annual averages per commune ranged between 72 million CFA (≈ 129,000 USD) and 77 million CFA (≈ 137,000 USD), while the corresponding medians show a relatively balanced distribution, with most communes receiving amounts close to the average. Over the full 2021–2024 period, half of the communes received a total of at least 179.6 million CFA (≈ 320,000 USD) in DCP allocations. This indicates that the overall distribution was relatively favorable, with a significant level of support provided to a majority of the beneficiary communes.

Table 1: Descriptive Statistics of DCP Transfers to Beneficiary Communes (2021–2024)

	Total DCP Amount	Mean	Min	Max	Median
2021	2,725,000,000	26,715,686	0	44,880,473	29,450,298
2022	0	0	0	0	0
2023	7,879,820,000	77,253,137	0	140,116,490	78,306,713
2024	7,396,330,931	72,513,048	0	120,091,817	74,929,319
Total	18,001,150,931	176,481,872	0	140,116,490	179,560,513

Notes: This table presents descriptive statistics of DCP transfers to beneficiary communes over the period 2021-2024. All amounts are in local currency units.

Beyond absolute amounts, it is essential to understand what these grants concretely represent within the budgetary structure of the communes. More than a simple financial contribution, the DCP has constituted—for many local governments—the main, if not the only, source of funding for local public investment. As shown in Figure 2, the DCP accounted on average for 75% of communes' investment capacity, 95% of their investment expenditures, and nearly all the state's investment subsidies.[7] In certain communes, the DCP made up virtually the entirety of their investment capacity and the majority of investment transfers received from the state over the

6. At the time of the study, the program was nearing completion, with some transfers still pending due to disbursement delays.

7. When calculating these various ratios by year, the 2023 figure exceeds 100%, reaching nearly 200% for certain aspects of investment. This indicates that the grants received through the DCP during this period were significantly higher than the actual investment capacity, the mobilized subsidies, or the executed expenditures. This gap reflects not only delays in disbursements but also in the receipt of funds from the state. These findings were confirmed by the PDREAS coordinator and the financial manager of the commune of Bangassi, who reported similar challenges in the management, mobilization, and use of funds related to local investment.

period. This was particularly evident in conflict-affected communes such as Abeibara, Adjelhoc, Bangassi, Gao, and Timbuktu, among others. These are also the communes that face the greatest difficulties in effectively mobilizing the DCP allocations to finance their investment expenditures.

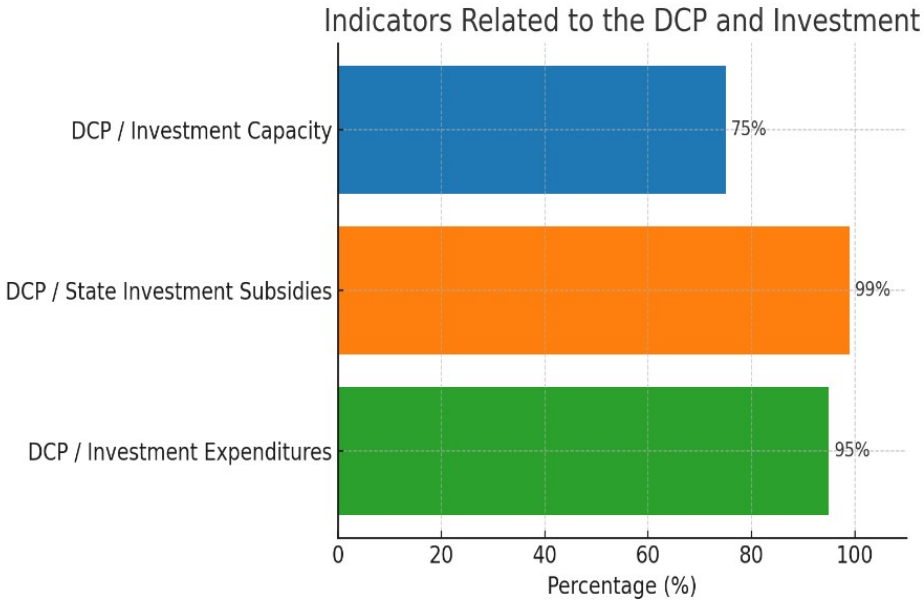


Figure 2: Comparative Ratios of the DCP Relative to Local Investment Capacity, State Investment Subsidies, and Executed Investment Expenditures (2021–2024)

PDREAS has established itself as a central lever for local investment, including outside conflict-affected areas. In fact, even in certain communes relatively spared by conflict, such as Narena, Youri, and Zantièbougou, the DCP at times represented the entirety of local investment during the period. In a context of shrinking public resources and the withdrawal of some technical and financial partners, PDREAS has become one of the few operational channels directly supporting productive investments at the local level.

This situation highlights a substitution effect, whereby the DCP tends to replace traditional national subsidies, thereby reducing both the incentive and the capacity of the state to intervene. The projects financed through the DCP have spanned a wide range of sectors: the rehabilitation or construction of classrooms, schools, community health centers (CSCOM), boreholes, vaccination centers, water infrastructure, and rural roads. These investments have often filled the gaps left by other funding sources and responded to critical needs expressed by the population.

Based on a two-level analysis, this study aims to assess the extent to which the PDREAS program has contributed to strengthening local governance, improving budget management in communes, and expanding access to essential services. First, the analysis will examine the effects of the Performance-Based Conditional Grants (DCP) on the investment capacity of communes by looking at achievements in the areas of educational, health, and agricultural infrastructure. It

will also evaluate the induced effects on budgetary efficiency, particularly in terms of execution rates, the functional allocation of expenditures, and the diversification of local revenues. In a second phase, the study will explore the program's effects on citizens' perceptions of public action, focusing on satisfaction with service delivery and trust in both local and national institutions. By combining budgetary data and opinion surveys, the study offers an integrated view of the effects of PDREAS, both in terms of the performance of local governments and in shaping perceptions of institutional legitimacy at the local and national levels.

PDREAS represents a unique experience of performance-based decentralization, both in Mali and across the subregion. This study seeks to capture the lessons learned, document its impacts, and formulate recommendations to ensure its sustainability and potential scale-up. The insights gained may also contribute to ongoing discussions on the reform of the fiscal transfer system, the territorialization of public policies, and the broader implementation of conditional grants across all communes in the country.

4 Empirical Strategy and Results

4.1 The Effect of PDREAS on Budgetary Capacity and Local Investment

The objective of this empirical analysis is to assess the extent to which the Performance-Based Conditional Grant (DCP) allocated under the PDREAS program has influenced local management and the structure of public goods expenditures. More specifically, the aim is to examine whether intercommunal and temporal variations in the amounts received are associated with significant differences in budgetary performance, namely execution rates, investment efforts, and sectoral trade-offs (education, health, electricity, water, agriculture). We estimate a reduced-form relationship between the amount of DCP received and various indicators of communal performance (see Appendix 7.2, Equation 1). The results are presented in Tables 2 and 3, which analyze the effects of the DCP on sectoral expenditures and on key budgetary and fiscal indicators, respectively.

The results presented in Table 2 highlight a differentiated effect of the DCP on the public spending priorities of beneficiary communes. It is first observed that expenditures on education, health, water, and agriculture are positively associated with the amount of DCP received. This reflects a reorientation of local budgets toward sectors with high social and visible impact, in line with the objectives of the PDREAS program and local development goals. The positive result in education is particularly striking. The coefficients of 0.11 for health and 0.16 for agriculture, both statistically significant, indicate that the DCP contributes to strengthening investment in healthcare systems and rural production. In contrast, the observed effect on infrastructure spending, although negative, is not statistically significant. This may be explained by the more rigid, complex, and delayed nature of infrastructure projects, which require procurement procedures, technical studies,

and multi-year planning.

Table 2: Effect of the DCP on Social Spending (Education and Health) and Infrastructure

	(1) Education	(2) Health	(3) Agriculture	(4) Infrastructure
DCP	0.35***	0.11**	0.16*	−0.01
	(0.017)	(0.05)	(0.09)	(0.07)
Commune Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
R^2	0.015	0.045	0.061	0.029
N	387	387	387	387

Notes: Robust standard errors clustered at the commune level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All regressions include commune and year fixed effects. The DCP variable and dependent variables are in logarithmic form.

Table 3 provides an overview of the effect of the DCP on the budgetary trajectory of communes. The results indicate that communes receiving higher amounts of DCP tend to invest more, while also improving their fiscal health and capacity to mobilize resources. Indeed, the positive relationship between the DCP and investment expenditures, with a statistically significant coefficient of 1.24, highlights a crowding-in effect. This trend may reflect a budgetary security effect : communes, assured of receiving conditional support, feel more confident to initiate or scale up investment projects they might not have undertaken otherwise. The incentive framework, combined with greater visibility over available resources, encourages more proactive and strategic behavior. The effect on the budget balance, with a high coefficient of 2.6, confirms an overall improvement in financial equilibrium, reflecting a greater ability of communes to plan, control, and adjust their expenditures. Similarly, the positive association between the DCP and tax revenues (0.20) points to a dynamic of increased accountability, with communes mobilizing more of their own-source revenues. Finally, the effect on the budget execution rate (0.8) reflects a growing competence in public resource management.

Table 3: Impact of the DCP on Local Investment, Tax Revenues, and the Budgetary Performance of Communes

	(1) Investment Expenditures	(2) Budget Balance	(3) Tax Revenues	(4) Budget Execution Rate
DCP	1.24***	2.26***	0.20***	0.8***
	(0.012)	(0.015)	(0.006)	(0.04)
Commune Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
R^2	0.015	0.045	0.061	0.022
N	387	387	387	387

Notes: Robust standard errors clustered at the commune level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All regressions include commune and year fixed effects. The DCP variable and dependent variables are in logarithmic form.

Our primary challenge for identification is the potential endogeneity of our treatment variable (DCP allocations). For example, central authorities may allocate more DCP resources to better-performing communes based on past performance, or alternatively prioritize underperforming or fragile areas for corrective support. In both cases, the allocation process is not exogenous and may be correlated with unobserved determinants of local investment performance, such as administrative capacity, political alignment, or local leadership quality. This creates a risk of biased estimates. While our empirical strategy includes commune fixed effects, which control for time-invariant characteristics such as geography, baseline capacity, or historical governance quality, and year fixed effects to absorb national-level shocks (e.g., political transitions, disbursement delays) that improve validity, they cannot fully address potential time-varying unobservables. As a robustness check, we re-estimate Table 2 and selected outcomes of Table 3 including the budget execution rate as an additional control variable, used here as a proxy for time-varying administrative capacity. As expected, the inclusion of this control slightly reduces the magnitude of the estimated DCP coefficients, indicating that part of the initial effect was correlated with variations in local financial management capacity. However, the results remain robust and statistically significant, suggesting that our fixed effects specification captures a substantial share of the underlying unobserved heterogeneity (see Table 9).

While this robustness exercise strengthens the internal validity of our findings, some limitations remain. Measurement errors may affect reported budget figures due to heterogeneous accounting practices or reporting delays. Moreover, despite improved robustness when controlling for budget execution, the non-random allocation of DCP still raises endogeneity concerns. A difference-

in-differences (DiD) strategy exploiting before-after variation could help address these issues by comparing performance trends between years without DCP and with DCP, while accounting for within-commune dynamics. However, the absence of a comparable group of untreated communes currently limits strict causal identification. Extending the analysis to include non-beneficiary communes would provide a stronger quasi-experimental framework for assessing program impacts.

4.2 Perception Analysis: Measuring the Effect of PDREAS on Citizen Satisfaction and Trust in Local and National Institutions

This analysis draws on Afrobarometer survey data collected before and after the implementation of the program to assess the effects of PDREAS on the quality of public services (education, health, water, employment) and on trust in both local and national authorities in the 102 participating communes. In the first stage, we examine Afrobarometer opinion questions regarding the public's assessment of government performance in six key sectors: (i) improving the living conditions of the poor, (ii) job creation, (iii) access to basic healthcare services, (iv) satisfaction of educational needs, (v) provision of water and sanitation services, and (vi) maintenance of road infrastructure. Responses, originally recorded on a four-level Likert scale ranging from "very bad" to "very good," are recoded into a binary format to distinguish between satisfied respondents ("fairly good" or "very good") and those who are not.

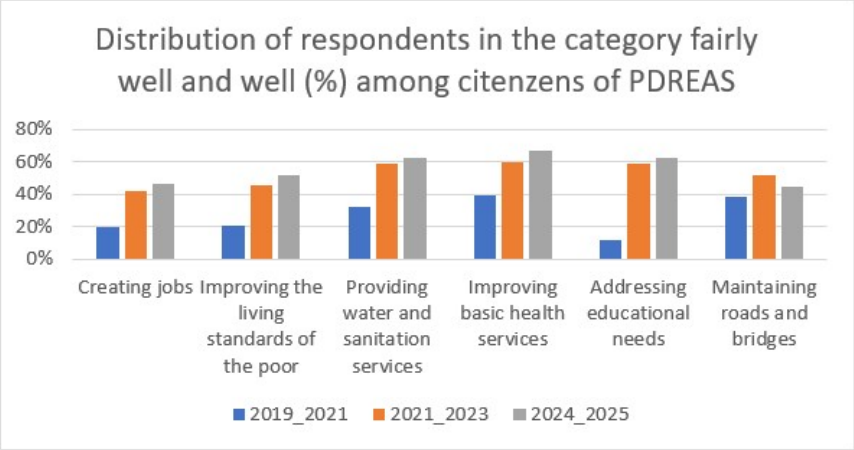


Figure 3: Trends in Citizens' Perceptions of Public Service Quality in PDREAS Communes (2019–2025)

Figure 3 shows the distribution of respondents who believe that certain public services are managed "fairly well" or "very well" by the government in PDREAS intervention areas. A general upward trend is observed across all sectors between the pre-program period (2019–2021) and the phases following program implementation (2021–2023 and 2024–2025). Before the program was launched, citizen perceptions were relatively low, particularly regarding job creation (only 20%

favorable responses) and the fulfillment of educational needs (11%). However, after the deployment of PDREAS, approval rates rose significantly—reaching 42% and then 46% for employment, and 59% then 63% for education.

Similar improvements are observed in the areas of access to water and sanitation (rising from 32% to 63%) and basic healthcare services (from 40% to 67%). These trends suggest that PDREAS interventions may have contributed to tangible improvements in the delivery of public services, particularly in key sectors. In contrast, road and bridge maintenance stands out as an exception. After an initial increase (from 39% to 52%), approval dropped to 44% in the most recent period, which may reflect ongoing challenges or unmet expectations in the infrastructure sector.

The quality of local governance is also examined from several angles. It is based in particular on three dimensions derived from citizens' perceptions: the perceived effectiveness of local elected officials in their roles, their responsiveness to citizens' concerns, and the level of trust placed in local authorities. The latter is measured through the question: "How much do you trust the following institutions, or have you not heard enough about them to say? Your local council / mayor." These indicators provide valuable insight into the ability of local governments to inspire trust, act transparently, and respond effectively to citizens' expectations, all of which are core objectives of the PDREAS program philosophy.

Figure 4 shows the evolution of citizens' perceptions of local institutions between two periods: 2019–2021 (before PDREAS) and 2021–2023 (after implementation). A marked improvement is observed across all three measured dimensions. The proportion of respondents who believe that local councillors are fulfilling their roles well rose from 47% to 65%, indicating a perceived improvement in the quality of local public action. Likewise, the share of citizens who feel that local officials listen to their concerns increased from 32% to 48%. Finally, trust in the elected municipal council rose significantly, from 33% to 56%. These developments suggest that PDREAS has contributed to strengthening the legitimacy and responsiveness of local authorities, in addition to its effects on budget management and public service delivery.

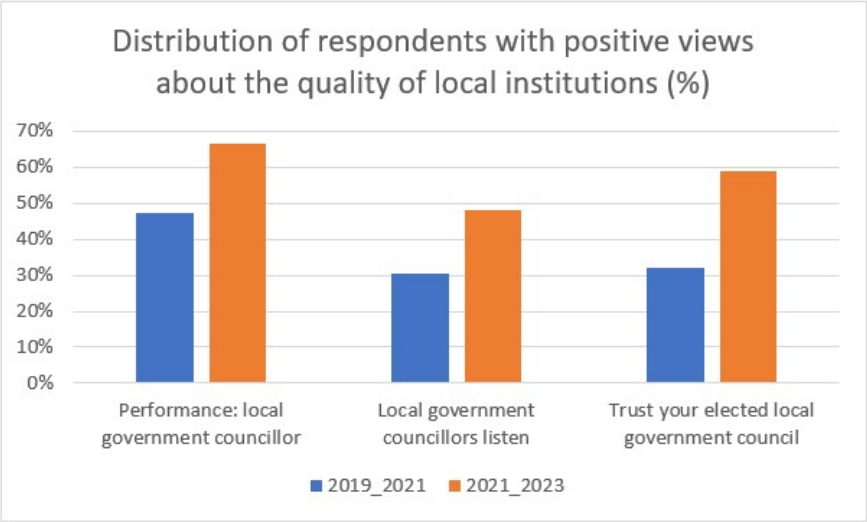


Figure 4: Trends in Trust and Perceptions of Local Institutions in PDREAS Communes (2019–2023)

This positive dynamic in local governance also appears to be accompanied by an improvement in trust in national institutions among citizens living in PDREAS beneficiary communes, as shown in Figure 5. A significant increase in trust in the presidency is observed, rising from around 50% to nearly 90% of respondents stating they have "some" or "a lot" of trust. Trust in the parliament also shows a notable rise, from 35% to over 65%. Trust in the ruling party, while increasing, remains more moderate. This positive trend may reflect, at least in part, the indirect effects of the PDREAS program, through improved quality of local public services and strengthened proximity governance. It suggests that beyond its immediate impacts on local governments, the program may also contribute to restoring institutional legitimacy at the national level.

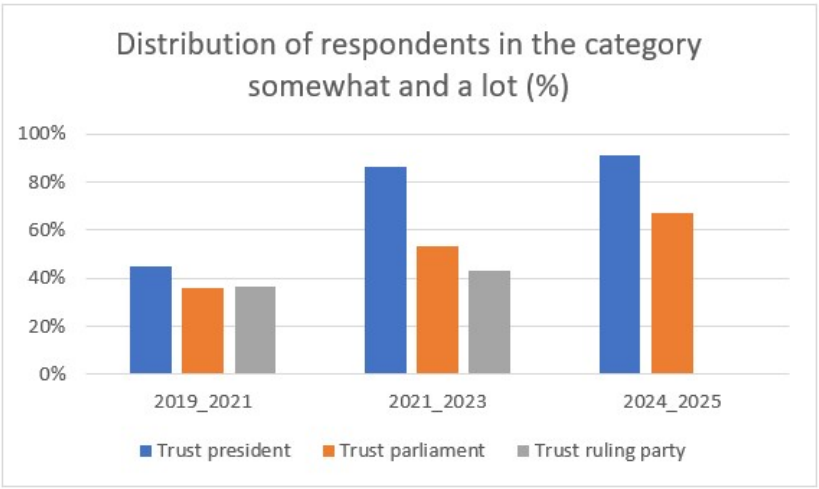


Figure 5: Trends in Trust in Central Institutions in PDREAS Communes (2019–2025)

Overall, the data reveal a marked improvement in citizen perceptions following the implementation of PDREAS. This positive dynamic is evident in the perceived quality of public services, in the legitimacy of local institutions, and, though to a lesser extent, in trust toward central authorities.

To more precisely assess the impact of PDREAS on citizens' attitudes and perceptions regarding public services, local institutions, and trust in national institutions, we employ an identification strategy based on a difference-in-differences (DiD) approach.[8] The results of the estimation of Equation (2) are presented in Tables (4)–(6).

Table 4 indicates that the implementation of the PDREAS program contributed to improving citizens' perceptions of the quality of local public services in several key sectors. The coefficients associated with the interaction between the DCP and the post-2021 period (DCP × Post) are positive and statistically significant for health, education, and access to clean water and sanitation, suggesting a positive effect of the program on beneficiary satisfaction.

Table 4: Difference-in-Differences Estimates of the DCP's Effect on Public Service Perception

	(1) Water & Sanitation	(2) Health	(3) Education	(4) Jobs	(5) Roads & Bridges
DCP*Post	**0.014*** **	**0.026*** **	**0.010*** **	0.0460	0.012
	(0.012)	**(0.015)**	**(0.006)**	(0.518)	(0.015)
Education level	−0.092***	−0.092***	−0.092***	−0.093***	−0.092***
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
Level of violence	−0.072**	−0.078**	−0.072**	−0.062**	−0.072**
	(0.030)	(0.030)	(0.030)	(0.030)	(0.030)
Gender (female)	0.017***	0.017**	0.017***	0.017***	0.017***
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)
Commune Fixed Effects	Yes	Yes	Yes	Yes	Yes
Round FE	Yes	Yes	Yes	Yes	Yes
R^2	0.165	0.165	0.165	0.165	0.165
N	7085	7085	7085	7085	7085

Notes: Robust standard errors clustered at the commune level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All regressions include commune and round fixed effects.

More specifically, the effect is particularly strong in the health sector, where the likelihood that an individual positively evaluates the management of healthcare services by the local government increases by 2.6 percentage points in communes that received the DCP after the program's launch. Similarly, satisfaction with water supply and sanitation services improves significantly, with an estimated effect of 1.4 points. Although more modest, the effect on education remains statistically significant, with a 1 percentage point increase. These results are consistent with the view that the additional budgetary resources provided through the DCP may have contributed to tangible

8. See Appendix 7.3.

improvements in socially impactful sectors, in line with the original priorities of the PDREAS program. These results may appear surprising given that the DCP financed only capital investments. However, a close analysis of the projects financed through the DCP in 2021, 2023, and 2024 provides insights that help explain the observed positive associations between DCP and local satisfaction. In many communes, DCP-funded projects involved the rehabilitation or upgrading of existing facilities, such as CSCOMs in Taboye and Talatayé or classrooms in Mopti and Fakala, which were already staffed or quickly reoccupied. In several cases, infrastructure delivery was complemented by basic equipment (e.g., medical kits, desks, or latrines), allowing services to become immediately more functional. In Sokoura, for example, school latrines and furniture contributed to better learning conditions, while in Niono and Sefeto Ouest, fencing and water points improved hygiene and security. By targeting relatively simple yet highly visible improvements, such as the functionality of health centers, the quality of school environments, or access to safe water and sanitation, these investments had a direct and immediate impact on the living conditions of local populations. This likely explains why citizens reported higher satisfaction with these services in the communes that benefited from DCP-funded projects.[9]

In contrast, the program's effect on job creation, while positive (4.6 percentage points), is not statistically significant. This may be due to the fact that PDREAS interventions did not directly target employment generation, or that labor market impacts require a longer time horizon to become observable. Similarly, no significant effect is found regarding perceptions of road and bridge maintenance. This result highlights the implementation challenges associated with large-scale infrastructure projects, which are often subject to longer timelines and local technical constraints.

Table 5 complements this analysis by focusing on citizens' perceptions of the quality of local institutions, particularly municipal councils. The results show that the PDREAS program had a differentiated impact on various dimensions of local governance. While the effects on perceived performance of elected officials and their responsiveness are positive but not statistically significant, the effect on trust in local institutions is both strong and significant. The coefficient of 0.184 indicates that, all else being equal, citizens living in a commune that benefited from the program are 18 percentage points more likely to trust their municipal council after the introduction of the DCP. This finding is particularly important in the context of a state undergoing reconstruction, where the consolidation of local governance is a key factor in political and social stabilization.

Extending the analysis of institutional trust beyond the local level, we now examine the extent to which the PDREAS program may have influenced citizens' perceptions of national and/or

9. Another plausible explanation for the observed effects could be a "signaling effect". Indeed, beyond functional improvements, the visible construction or rehabilitation of service infrastructure may be perceived by citizens as a tangible sign of renewed government presence and responsiveness. This perception alone can positively influence satisfaction and trust, even before services are fully operational. In fragile contexts, where state action has historically been delayed or absent, such visibility can strengthen expectations and reinforce the social contract.

Table 5: DCP and the Quality of Local Governance

	(1) Councillor Performance	(2) Councillors' Responsiveness	(3) Trust in Local Council
DCP*Post	0.127	0.131	**0.184*** **
	(0.106)	(0.107)	**(0.103)**
Education level	−1.792**	−1.766**	−2.381***
	(0.787)	(0.786)	(0.772)
Level of violence	−0.019	−0.018	−0.043
	(0.022)	(0.022)	(0.028)
Gender (female)	0.0330***	0.0343***	0.0456***
	(0.0051)	(0.0050)	(0.0052)
Commune Fixed Effects	Yes	Yes	Yes
Round FE	Yes	Yes	Yes
R^2	0.120	0.123	0.120
N	5883	5883	5883

Notes: Robust standard errors clustered at the commune level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All regressions include commune and round fixed effects.

central political institutions in beneficiary communes. Table 6 presents the results of the PDREAS program's effect on reported trust in three national institutions: the presidency, the national assembly, and the ruling party. The estimated coefficients indicate positive effects, with a statistically significant impact of the DCP on trust in the president (9.1 percentage points) and in the national assembly (7.3 percentage points). In contrast, the effect on trust in the ruling party remains very limited and statistically insignificant. These findings suggest that although the program is focused on local governance, it may have contributed to enhancing the perceived legitimacy of national institutions.

Table 6: DCP and Trust in National Institutions

	(1) Trust in President	(2) Trust in Assembly	(3) Trust in Ruling Party
DCP*Post	**0.091*** **	**0.073*** **	0.011
	(0.032)	**(0.032)**	(0.007)
Education level	−1.435**	−1.652**	−2.045***
	(0.690)	(0.730)	(0.690)
Level of violence	−0.030*	−0.077*	−0.043
	(0.017)	(0.032)	(0.028)
Gender (female)	0.0291***	0.0327***	0.0412***
	(0.0048)	(0.0046)	(0.0050)
Commune Fixed Effects	Yes	Yes	Yes
Round FE	Yes	Yes	Yes
R^2	0.128	0.132	0.115
N	7085	7085	5883

Notes: Robust standard errors clustered at the commune level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All regressions include commune and round fixed effects.

Finally, the control variables confirm several empirical regularities. A higher level of education is systematically associated with more critical evaluations of public services and authorities, both local and national, while women tend to express more positive views. In contrast, the intensity of armed violence in the commune (measured using ACLED data) is significantly correlated with lower satisfaction across all three dimensions analyzed: the quality of public services, local governance, and trust in central institutions. This finding underscores the disruptive effect of conflict on the relationship between citizens and institutions.

4.3 Convergence between Administrative Performance and Citizen Perception

The combined analysis of Sections 4.1 and 4.2 reveals how the program's measurable outcomes are clearly reflected in the satisfaction and trust expressed by citizens. This is especially evident in social sectors such as education, health, and agriculture, where increased investments and expenditures correspond closely with noticeable improvements in citizen satisfaction. Residents appear to directly perceive the benefits of improved budget management, particularly in health care services and access to clean water, reinforcing the notion that the resources allocated through the program have been effectively mobilized to serve the community.

This coherence also extends to the overall strengthening of local governance. Improved budget execution and more rigorous financial management are reflected in increased citizen trust in local elected officials. The results thus suggest that the enhancement of administrative capacities induced by the program has contributed positively to rebuilding the trust relationship between local governments and their constituents.

These converging dynamics between technical performance and citizen perception lend support to the transformative potential of the DCP. By leveraging targeted budgetary incentives, the program appears to have contributed to improvements in the efficiency of local spending as well as to a strengthening of the legitimacy of communal authorities in the eyes of the population. In a fragile institutional context, it serves as a credible lever for reshaping the relationship between the local state and its citizens. These findings support consideration of the consolidation and gradual expansion of the mechanism, provided that its performance requirements are maintained and the management capacities of communes are further strengthened. Malian authorities would thus benefit from viewing the DCP as a strategic tool for local governance, capable of aligning tangible results with social expectations.

5 A Scenario-Based Cost-Effectiveness Illustration

This section presents a scenario-based, assumption-driven illustration of the potential fiscal returns associated with the institutional trust gains documented in Section 4.2. The exercise is not a forecast: it is a transparent simulation of what the documented trust improvements could imply for tax compliance and revenue mobilization, conditional on a specific set of behavioral and fiscal assumptions drawn from the recent literature. The reasoning is laid out step by step (Section 5.1), applied to a baseline scenario (Section 5.2), and then probed for sensitivity to its main parameters (Section 5.3).

5.1 Construction of the Scenario

The simulation proceeds in five explicit steps. Each step rests on a specific assumption that we make visible so that readers can substitute their own preferred values and recompute the implied returns.

Step 1 - Trust gain. The difference-in-differences estimates in Section 4.2 indicate a +18.4 pp increase in trust in municipal councils, +9.1 pp in the presidency, and +7.3 pp in parliament. The baseline aggregates the local and presidential effects (+27.5 pp), conservatively setting aside the parliamentary effect to limit potential double-counting across correlated political institutions.

Step 2 - Translating trust into compliance. We rely on Besley and Dray (2024), who estimate an elasticity of approximately 2.15 points of voluntary tax compliance per point of institutional trust.

The baseline trust gain therefore implies a compliance index improvement of $27.5 \times 2.15 \approx 59.1$ points.

Step 3 - From compliance to revenue. Not all gains in the compliance index translate one-for-one into collected revenue, given Mali's large informal sector and limited administrative reach. We assume that 2 to 5 percent of the compliance index gain materializes as effective revenue, yielding an effective collection increase of approximately 1.18 to 2.96 percent.

Step 4 - Application to the tax base. With a 2024 nominal GDP of USD 26.59 billion and a tax-to-GDP ratio of 18 percent, baseline annual tax revenue stands at approximately USD 4.79 billion. Applying Step 3 yields illustrative additional annual tax revenues of approximately USD 57 million to USD 141 million.

Step 5 - Comparison with program cost. PDREAS was funded with an initial envelope of USD 50 million. To compare an annual fiscal flow with the program cost on a consistent basis, we interpret USD 50 million as the recurrent annual envelope that an institutionalized PDREAS would require to operate at current scale. This is the most direct flow-to-flow comparison; alternative readings are addressed in Section 5.3.

5.2 Baseline Scenario

Combining the five steps above yields illustrative annual fiscal returns of USD 57 million to USD 141 million against an assumed recurrent annual program cost of USD 50 million, corresponding to a cost-coverage ratio of approximately 1.1 to 2.8 and a net annual fiscal gain of USD 7 million to USD 92 million. Even at the lower end, the program would broadly cover its annual recurrent cost; at the upper end, it would generate a fiscal return of close to three times its cost.

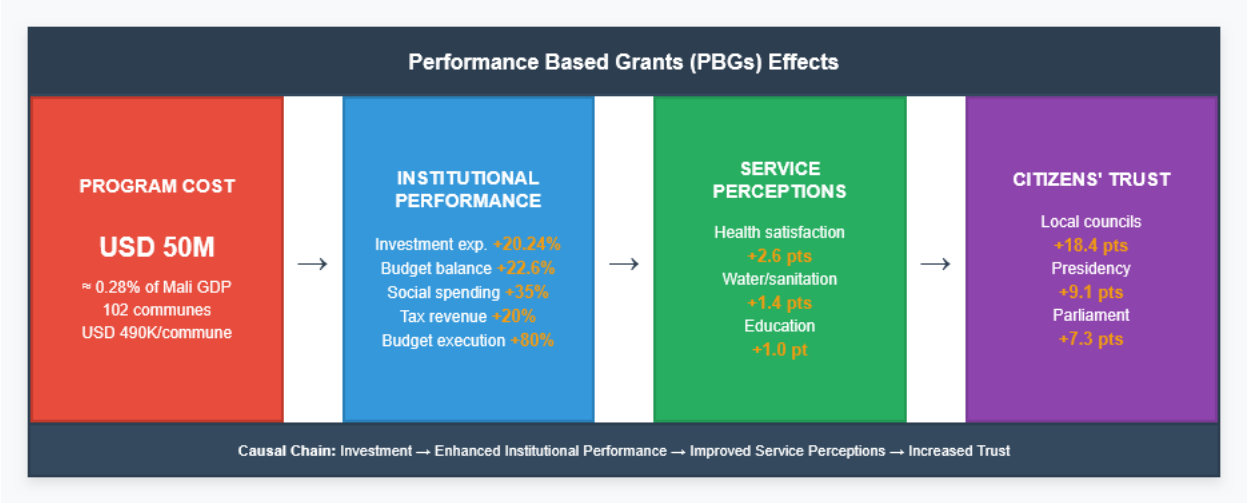


Figure 6: PDREAS Program Impact Chain - From Investment to Institutional Trust

These figures should be read alongside the broader institutional value of trust gains, which the

academic literature frequently associates with a stabilizing role against political violence in fragile contexts. Such non-fiscal benefits are not captured in the figures above.

5.3 Sensitivity to Key Parameters

The headline ratio of 1.1 to 2.8 rests on the joint validity of all five assumptions described in Section 5.1. To make this sensitivity transparent, Table 7 presents three illustrative scenarios that vary the two parameters most directly affecting the result - the aggregation of trust gains (Step 1) and the pass-through from compliance to revenue (Step 3) - relative to the baseline.

Table 7: Sensitivity of Illustrative Annual Fiscal Returns to Key Assumptions

Scenario	**Trust gain**	**Pass-through**	**Annual return (USD M)**	**Cost-coverage ratio**
Lower	Local only (+18.4 pp)	1–2.5%	19–47	0.4–0.9x
Baseline	**Local + Presidency (+27.5 pp)**	**2–5%**	**57–141**	**1.1–2.8x**
Upper	All three levels (+34.8 pp)	2–5%	72–179	1.4–3.6x

Notes: All scenarios apply the trust-to-compliance elasticity of 2.15 from Besley and Dray (2024) to a 2024 baseline tax revenue of USD 4.79 billion (18% of nominal GDP of USD 26.59 billion). The cost-coverage ratio is computed against an assumed recurrent annual program cost of USD 50 million. The baseline (in bold) corresponds to the scenario discussed in Section 5.2. The Lower scenario halves the pass-through assumption and restricts the trust gain to the local effect. The Upper scenario aggregates all three documented trust effects. None of these figures should be interpreted as forecasts.

Two additional sources of uncertainty deserve explicit mention. *First*, the trust-to-compliance elasticity of 2.15 is drawn from a context that differs from Mali's; a 25% reduction of this parameter would proportionally reduce the implied returns under all three scenarios. *Second*, the entire exercise rests on the assumption that the trust gains are causally attributable to PDREAS rather than to concurrent political shocks, notably the 2020–2021 political transitions. Our difference-in-differences design mitigates this concern under parallel-trends assumptions but cannot fully establish causality, particularly for spillover effects on national institutions; should a substantial share of the documented gains be attributable to other factors, the implied fiscal returns would shrink proportionally.

Taken together, these sensitivities suggest that the headline ratio of 1.1 to 2.8 should be read as the central estimate of a plausible range that spans from below break-even at the most conservative configuration to a multiple of three to four times the annual cost at the upper end. The encouraging early results of the pilot phase, combined with this range of plausible fiscal returns, provide supportive evidence for considering further expansion of PDREAS, alongside the complementary efforts discussed in Section 6.

6 Conclusion, Discussion, and Policy Considerations

The findings of this study are consistent with the view that PDREAS, through its DCP, may serve as a promising instrument for improving the quality of local governance and strengthening the delivery of basic public services. In communes where budget execution improved, the DCP was associated with measurable increases in social investments (in education, health, and water), as well as with observed improvements in citizens' trust in local institutions, suggesting potential to support institutional transformation in fragile contexts.

These findings underscore the strategic importance of institutional trust as a pivotal element for sustainable state-building, conflict prevention, and democratic consolidation. Leveraging the empirical insights of Besley and Dray (2024), the analysis also identifies significant fiscal potential associated with increased institutional trust. Specifically, scenario-based estimates suggest that observed improvements in trust levels under PDREAS could be associated with additional annual fiscal revenues on the order of USD 57 million to USD 141 million, approximately 1.1 to 2.8 times an assumed recurrent annual program cost of USD 50 million, pointing to the potential financial viability of this approach under the stated assumptions, and subject to the sensitivities discussed in Section 5.3.

Beyond its financial implications, the program appears to have contributed to a strengthening of the legitimacy of the state at the local level. Afrobarometer data show that the implementation of PDREAS is associated with a marked improvement in citizens' perceptions, not only regarding the quality of social services (education, health, water, employment) and the performance of local authorities (responsiveness, competence, trust in elected officials), but also, to some extent, with trust in central institutions (the presidency, parliament, and the ruling party). This link between financial incentives, institutional performance, and public trust is all the more relevant in a Malian context marked by persistent security challenges and widespread distrust of institutions.

Despite these encouraging early results, certain anticipated outcomes such as job creation and road infrastructure maintenance did not fully materialize, likely reflecting inherent complexities in these sectors and disruptions due to political instability and the COVID-19 pandemic. Additionally, limitations in survey data, particularly the absence of detailed information on local officials, have constrained a more nuanced understanding of variability in commune performance. Addressing these data gaps through more granular data collection, and complementing quantitative analyses with qualitative methods such as interviews or focus groups, would be a useful priority for future research.

The remainder of this section outlines a set of policy options and considerations that may inform reflection on the program's future. These are presented as possible directions worth weighing, rather than as prescriptive recommendations: their appropriateness will ultimately depend on the political timing, administrative capacity, and complementary reforms that surround any decision

to scale or institutionalize the PDREAS approach.

First option: gradual evolution toward a more impact-based evaluation framework. Current evaluation criteria primarily reward administrative compliance, such as reporting deadlines and digitalization, with service delivery quality accounting for approximately 20% of the performance assessment. Reweighting the criteria to give greater prominence to results-oriented indicators, such as perceived quality of public services, user satisfaction, literacy rates, or health outcomes, could plausibly amplify the impact of the DCP. International experience offers some indication that such approaches can be productive: Rwanda and Uganda, for example, have incorporated service quality and citizen satisfaction metrics into performance-based evaluations, with reported gains in service outcomes and accountability. The transferability of these experiences to the Malian context would itself merit careful consideration.

Second option: strengthening community participation mechanisms. A second avenue worth considering involves strengthening mechanisms for community participation, with a view to ensuring that investments more closely reflect community priorities and foster local ownership. Evidence from other fragile contexts, such as that summarized by Bardhan and Mookherjee (2006), suggests that meaningful citizen involvement may help reduce elite capture, support legitimacy, and contribute to program sustainability, though the channels through which these effects operate vary across institutional settings.

Third option: possible institutionalization and gradual scaling within the national fiscal framework. A third area worth considering concerns the possible institutionalization and gradual scaling of the PDREAS approach within Mali's national fiscal framework. The recent adoption of the 2023 Local Government Code provides a legal foundation that could support the integration of performance-based grants into the national system. International experience, including from Tanzania and Uganda, where performance-based grants have been progressively institutionalized, suggests that formalizing the DCP through dedicated budget classifications and regulatory frameworks could improve the transparency, predictability, and sustainability of funding, while recognizing that the timing and modalities of any such step would need to align with broader fiscal reform priorities.

Fourth option: geographic targeting of any future expansion. A fourth consideration concerns the geographic targeting of any future expansion. A gradual, adaptive approach, prioritizing fragile communes that nonetheless display sufficient governance capacity, and pairing expansion with targeted capacity-building initiatives, could help support consistent progress across diverse governance environments and limit the risk of widening regional disparities. Comparable approaches in other contexts, such as Uganda, offer some illustrative reference points, though the specific design choices would need to reflect Mali's own territorial realities.

Overall, this study provides supportive evidence that PDREAS may constitute a promising

governance instrument with potentially meaningful financial and institutional implications under stated assumptions. Should authorities and partners decide to pursue further implementation and gradual scaling, the options outlined above, including a stronger orientation toward results-based criteria and community engagement, could plausibly contribute to reinforcing Mali's local governance structures, supporting public service delivery, and gradually rebuilding citizen trust in public institutions.

A useful avenue for future research would be to extend the analysis to a multi-country dimension. A comparative study including several countries that employ similar performance-based conditional grant mechanisms would allow for a deeper understanding of the contextual and institutional factors shaping program effectiveness. Extending the analysis to other Sub-Saharan African countries, such as Burkina Faso, Tanzania, and Uganda, could offer valuable comparative insights, helping to identify approaches that travel well across contexts and adaptations that fragile settings may require. Mixed-method approaches, integrating quantitative analyses with detailed qualitative assessments, would further help clarify the mechanisms through which local governance is strengthened and institutional trust is built.

7 Appendices

7.1 Abbreviations

7.2 Appendix to Section 3.1

7.2.1 Data

To analyze the effect of PDREAS on budgetary capacity and local investment, our primary variable of interest is the amount of the Performance-Based Conditional Grant (DCP) received by each commune under the PDREAS program between 2021 and 2024. These amounts were provided directly by the program coordination team, based on validated disbursements and annual assessments of commune performance. The DCP represents an investment transfer conditional on meeting a set of governance, budget management, and transparency criteria, and therefore serves as a strategic lever within the program's incentive framework. It is used here as the key explanatory variable in our analysis.

Our local performance variables (outcome variables) are constructed from an original administrative dataset covering the executed budget data of the 102 PDREAS beneficiary communes over the 2021–2024 period. This dataset, compiled from the annual management accounts produced by the Treasury, allows for a detailed reconstruction of all local financial flows—both revenues and expenditures—while distinguishing between operating and investment transactions. It constitutes a rare resource in the African context, due to its level of disaggregation, multi-year coverage, and

Table 8: List of Acronyms

Acronym	Definition
ACLED	Armed Conflict Location & Event Data
ANICT	National Local Government Investment Agency
BAR	Results-Based Budgeting
BVG	Office of the Auditor General
CROCSAD	Regional Development Action Steering, Coordination and Monitoring Committee
CSCOM	Community Health Center
DCP	Performance-Based Conditional Grant
DiD	Difference-in-Differences
DGCT	General Directorate of Local Governments
DGD	General Decentralization Grant
DRPSIA	Regional Directorates for Planning, Statistics, and Information Systems for Land Use and Population Planning
EA	Enumeration Area
FCV	Fragility, Conflict & Violence
FNACT	National Support Fund for Local Governments
IDA	International Development Association
IMF	International Monetary Fund
PACUM	Urban Local Government Support Project (IDA 4966)
PDESC	Economic, Social and Cultural Development Plan
PDREAS	State Resources Deployment Project for the Improvement of Services
PEFA	Public Expenditure and Financial Accountability Assessment

consistency with the mechanisms used to monitor local performance.

On the revenue side, the dataset distinguishes between own-source tax revenues (such as the local development tax, market fees, and land-use royalties), non-tax revenues (miscellaneous income, fines, rents), and, most importantly, transfers from the central government. Among these transfers, unconditional transfers such as the General Decentralization Grant (DGD) coexist with targeted transfers, including the DCP. These grants account for a substantial share of local resources in beneficiary communes: the DCP alone represents between 20% and 40% of total revenues in some years, underscoring its importance in shaping investment budgets.

Regarding expenditures, the dataset clearly distinguishes between operating expenditures (such as personnel costs and routine management) and investment expenditures (infrastructure, equipment, services). Investment spending is also broken down by functional sector—education, health, water, agriculture, and road infrastructure—which allows for a precise analysis of local governments' strategic priorities, and an empirical test of whether PDREAS contributed to reorienting local public spending toward social sectors. Based on the raw budget variables, we constructed the following indicators used in the empirical analysis. These indicators allow us to assess the program's effects on budget management capacity, the orientation of local expenditures, and the sustainability of municipal finances. Each indicator is defined to capture a specific dimension of budgetary performance:

- ▷ **Overall Budget Execution Rate:** This indicator measures the overall capacity of a commune to execute its budget. It is defined as the ratio between total actual expenditures (operating + investment) and the total approved budget at the end of the fiscal year. It reflects overall administrative performance and the financial absorption capacity of the local government.

- ▷ **Investment Expenditures:** This variable captures the total amount of investment spending executed by the commune. It helps assess the level of public investment mobilized during the implementation of the program.

- ▷ **Sectoral Composition of Investment:** These variables measure the distribution of investment expenditures across sectors:

  - **Education, health, water:** Reflects the social orientation of investments.
  - **General infrastructure:** Captures spending on roads, public buildings, and equipment.
  - **Agriculture:** Indicates the attention given to economic and rural development functions.

- ▷ **Budget Balance:** The budget balance is defined as the difference between total revenues collected and total expenditures executed. It is used to evaluate the financial equilibrium of the commune. A positive balance indicates surplus or prudent financial management, whereas a negative balance may reflect budgetary fragility or pressure on local resources.

The combination of functional, temporal, and spatial granularity makes this dataset a central analytical tool for our study. It enables us to link the amounts of DCP received to a series of indicators reflecting budgetary performance, expenditure structure, and political trade-offs made at the communal level.

7.2.2 Empirical Strategy

As previously mentioned, the objective of our empirical analysis is to assess the extent to which the Performance-Based Conditional Grant (DCP), allocated under the PDREAS program, has influenced the dynamics of local management and the investissement in public services. More specifically, we estimate the following econometric model:

$$Y_{ct} = \beta \times DCP_{ct} + \mu_c + \lambda_t + \varepsilon_{ct} \tag{1}$$

where Y_{ct} denotes the outcome variable observed for commune c in year t, DCP_{ct} represents the amount (or relative share) of the DCP received that year, μ_c is a fixed effect for each commune, λ_t is a time fixed effect (by year), and ε_{ct} is the idiosyncratic error term. This model allows us to control for time-invariant unobserved characteristics at the commune level (such as poverty levels, administrative capacity, geographic conditions, or historical governance quality), as well as for nationwide shocks affecting all communes at a given time (e.g., economic crises, government changes, pandemics, etc.).

However, this empirical framework has a major limitation in terms of causal identification: the DCP is not randomly assigned. The amount allocated depends both on the communes' performance across several criteria (such as revenue mobilization, transparency, and budget execution) and on an ex post evaluation process. As such, there is a risk of endogeneity: a better-managed commune is more likely to receive higher DCP allocations, which may bias the estimated effect of the program. Conversely, underperforming communes might be prioritized, potentially leading to a negative relationship between DCP and outcomes—even if that does not reflect a true program effect. In the meantime, and to mitigate potential bias within the current empirical framework restricted to treated communes, we adopt an alternative robustness check. Specifically, we re-estimate equation (2) by including the Budget Balance as an additional control variable, which serves as a proxy for time-varying local financial and administrative capacity. This allows us to test whether the observed effects of the DCP remain robust when controlling for unobserved heterogeneity linked to commune management quality.

As expected, adding the Budget Execution Rate as a control variable slightly reduces the magnitude of the estimated DCP coefficients on investment, budget balance, and tax revenues. This suggests that part of the initial effect was correlated with variations in local financial management capacity. However, the coefficients remain robust and statistically significant, confirming that the

Table 9: Effect of the DCP on Social Spending, Expenditures and Tax Revenues

	(1) Education	(2) Health	(3) Agriculture	(4) Investment Expenditures	(5) Tax Revenues
DCP	0.28**	0.09**	0.11**	1.39***	0.16***
	(0.016)	(0.05)	(0.09)	(0.015)	(0.07)
Commune Fixed Effects	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes
Execution Rate	Yes	Yes	Yes	Yes	Yes
R^2	0.028	0.055	0.067	0.032	0.075
N	387	387	387	387	387

Notes: Robust standard errors clustered at the commune level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. All regressions include commune and year fixed effects. The DCP variable and dependent variables are in logarithmic form.

DCP exerts an independent and positive effect on key dimensions of local budgetary performance.

While this robustness check strengthens the internal validity of our findings, several limitations must be acknowledged. In particular, measurement errors may affect reported budget figures, notably due to heterogeneous accounting practices or delays in financial reporting. Furthermore, although controlling for Budget Execution Rate reduces some endogeneity concerns, the non-random allocation of Performance-Based Conditional Grants (DCP) remains a potential source of bias. To address these challenges more rigorously, a difference-in-differences (DiD) econometric strategy, exploiting the temporal "before-after" variation, could be employed. This approach would compare the evolution of budgetary and governance performance indicators between years without DCP (notably 2022) and those with DCP (2021, 2023, 2024), while accounting for within-commune variation. However, given the absence of a comparable group of untreated communes, this analysis cannot support strict causal identification within the current framework. A promising extension would therefore be to include a group of non-beneficiary communes, allowing the construction of a credible control group and enabling a more robust quasi-experimental assessment of the program's effects.

7.3 Appendix to Section 3.2

7.3.1 Data

This empirical analysis draws on georeferenced Afrobarometer survey data covering the period from 2008 to 2025. Afrobarometer is a valuable and representative source of public opinion surveys across many African countries, providing detailed information on political attitudes, satisfaction with public services, and trust in local institutions. The surveys allow for the measurement of citizens' perceptions regarding the quality and accessibility of basic services—such as education,

health, water, and local infrastructure—as well as their level of trust in elected authorities at both local and national levels.

Each respondent is linked to a specific geographic unit—the Enumeration Area (EA)—for which GPS coordinates are available. This spatial information allows us to associate each respondent with a commune in Mali that either did or did not benefit from the PDREAS program. By cross-referencing the program implementation dates with the Afrobarometer data, we construct a treatment variable indicating whether a commune was exposed to the Performance-Based Conditional Grant (DCP) at the time of the survey. This matching strategy makes it possible to compare citizen perceptions between beneficiary and non-beneficiary communes, before and after the program's rollout, thereby enabling an assessment of the potential impact of PDREAS on satisfaction with public service delivery (education, health, water, etc.) and trust in local institutions (see, for example, Konté (2020) and Konté et al. (2021)).

Table 10: Distribution of Individuals by Location

	Number of Individuals	Percentage (%)
Non PDREAS	5,272	73.22
PDREAS	1,928	26.78
Total	7,200	100.00

7.3.2 Empirical Strategy

A To evaluate the impact of the Special Administrative Support Regions Development Program (PDREAS) on citizens' attitudes and perceptions toward public services and local institutions, we use an identification strategy based on a difference-in-differences (DiD) approach.

Our analysis relies on a subsample of Afrobarometer surveys, restricted to Malian communes potentially eligible for the program, and to survey waves conducted before (2008–2020) and after (2021–2025) the launch of PDREAS. Each respondent is geographically located based on their Enumeration Area (EA)—the census unit for which GPS coordinates are available. This spatial information allows us to link each individual to their respective commune and thus identify whether they reside in a commune that benefited from the program. By aligning the PDREAS implementation timeline with respondent locations, we define a treatment variable indicating effective exposure to the Performance-Based Conditional Grant (DCP) at the commune level, starting in 2021.

Moreover, to account for contextual shocks that may influence the results, we include a control variable for violence based on ACLED data. This variable measures the cumulative number of armed violence events by commune (excluding peaceful protests and non-violent demonstrations), including clashes, explosions, and attacks against civilians. It allows us to control for the potentially

disruptive effect of local conflict on service delivery and perceptions of institutions. The evaluation is based on the following specification :

$$Y_{ict} = \alpha + \delta \cdot (\text{DCP}_c \times \text{Post}_t) + \beta_1 \text{DCP}_c + \beta_2 \text{Post}_t + \gamma X_{ict} + \lambda Z_{ct} + \mu_c + \tau_t + \varepsilon_{ict} \tag{2}$$

Equation (2) is estimated using a linear probability model applied to a binary outcome variable, where Y_{ict} denotes the outcome of interest measuring the satisfaction of individual i, residing in commune c, in survey wave t, with respect to various aspects of local public action. On one hand, we include binary indicators reflecting citizens' evaluations of government performance in key service delivery sectors—education, health, access to water, and employment. These variables capture declared satisfaction with essential public services. On the other hand, we use two groups of indicators related to governance and institutional legitimacy. At the local level, we consider perceptions of the effectiveness of municipal councilors in carrying out their duties, their responsiveness to citizen concerns, and trust in elected municipal councils. At the central level, we include indicators measuring trust in major national institutions, particularly the president, the parliament, and the ruling party. By combining these different dimensions, the approach allows us to assess the extent to which PDREAS, through its conditional grants, contributes to strengthening citizen trust in public institutions—both local and national.

The variable DCP_c is a binary indicator equal to 1 if commune c received a grant under the PDREAS program. The term Post_t captures periods after 2021, the year the program was launched. The coefficient associated with the interaction term $(\text{DCP}_c \times \text{Post}_t)$, denoted δ, is our main parameter of interest: it measures the differential effect of the program on citizen perceptions in treated communes after its implementation.

To better account for observed heterogeneity across individuals and communes, we include a set of individual-level control variables X_{ict}, including gender, age, education level, employment status, and place of residence (urban or rural). In addition, we include a contextual variable (Z_{ct}), specifically a measure of the intensity of violence in each commune, based on ACLED data. This variable aggregates violent events recorded at the commune level—such as armed attacks, explosions, and clashes—while excluding peaceful protests. It is intended to capture the local security environment, which may influence both the provision of services and citizens' perceptions of public institutions.

Finally, our model includes commune fixed effects (μ_c) and time (survey wave) fixed effects (τ_t) to control for unobserved characteristics that are either constant over time within communes or common to all individuals surveyed in a given year. This specification is particularly important given the political context in Mali, where national-level shocks—such as the coups d'état in 2020 and 2021—may have influenced local perceptions and institutional functioning. The inclusion of time fixed effects helps absorb such shocks, reducing the risk of bias in estimating the association

between DCP exposure and individual outcomes. Estimation is performed using ordinary least squares (OLS), with robust standard errors clustered at the commune level to account for potential intra-group correlation in the error term (ε_{ict}).

This strategy makes it possible to isolate the specific effect of PDREAS on the outcomes of interest by comparing changes in citizen perceptions between treated and untreated communes, while controlling for overall trends and local specificities. It builds on recent work examining the impact of public policies on citizen attitudes, particularly studies on resource windfalls and the quality of local governance (Konte and Vincent, 2021, Konte, 2021). The key question here is whether a conditional transfer instrument such as the DCP can improve citizen trust and the perceived quality of public action in a context of institutional fragility.

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